A Feature-Based Approach to Panoramic Image Stitching Using SIFT and RANSAC

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Abstract

We present a feature-based approach for panoramic image stitching, utilizing Scale-Invariant Feature Transform (SIFT) and RANSAC for homography estimation. Tested on synthetic and real-world datasets, the method effectively handles various transformations and image imperfections, suggesting that while the current blending methods suffice for simple cases, enhancements like multi-band blending could improve results in complex scenarios.

1. Introduction

Panoramic image stitching is a well-known problem in computer vision, where multiple overlapping images are combined into a single, larger field of view. This process has applications ranging from aerial photography and virtual tours to robotic navigation and medical imaging. By stitching together two or more images taken from different angles or at different times, it is possible to obtain a comprehensive representation of a scene that may not be possible to capture in a single shot.

For this project, I chose the feature-based stitching approach because it is robust to moderate changes in viewpoint, scale and lighting. Specifically, I used the Scale-Invariant Feature Transform (SIFT) for keypoint detection and descriptor extraction, combined with the RANSAC algorithm for homography estimation. This choice reflects both the maturity of SIFT in the academic literature and its proven effectiveness in producing high-quality panoramic images. I implemented and tested the solution in Python (notebooks) using the OpenCV library because of its extensive support for feature detection, matching and geometric transformations.

2. Methodology

2.1 Feature Detection and Description

We employ the Scale-Invariant Feature Transform (SIFT) as our primary method for extracting local features. SIFT is known for its resilience to changes in scale, rotation, and moderate viewpoint alterations. Specifically, each input image (converted to grayscale) undergoes:

1. **Scale-space extrema detection:** I constructed a Gaussian pyramid and Difference-of-Gaussian images at multiple octaves to find keypoints that are maxima/minima.

2. **Keypoint localization and filtering:** Low-contrast points and edges are discarded, retaining only robust keypoints.

3. **Orientation assignment:** Each keypoint is assigned one or more orientations based on local image gradients, conferring rotation invariance.

4. **Descriptor computation:** Around each keypoint, I computed a 128-dimensional histogram-of-orientations descriptor, which is normalized to enhance robustness to intensity variations.

Since SIFT can generate hundreds or thousands of features per image, I optionally limited the maximum number of keypoints (e.g., to 2,000) to control computational load, retaining the top responses.

2.2 Feature Matching

Once the SIFT descriptors for both images are extracted, I continued to match them. I used a brute-force matching strategy for simplicity, implemented via OpenCV's `BFMatcher` class, although more efficient methods (e.g., FLANN-based matching) are also available.

Lowe's Ratio Test. To mitigate erroneous matches, I adopted Lowe's ratio test, in which each descriptor from the first image is matched with its two nearest neighbors in the second image's descriptor set. Let d_1 and d_2 be the distances to the best and second-best matches, respectively. I only accept the match if:

$$\frac{d_1}{d_2} < \alpha, \tag{1}$$

where α (typically $0.7 - 0.8$) is a user-defined threshold. This helps remove ambiguous matches and ensures greater distinctiveness of the retained keypoint correspondences.

2.3 Homography Estimation

After filtering the matches, I obtained a set of correspondences between 2D keypoints in the first and second images. These pairs of points, denoted $(x_{\text{img2}}, y_{\text{img2}})$ and $(x_{\text{img1}}, y_{\text{img1}})$, are used to estimate the 3×3 homography matrix H that maps coordinates in the second image to coordinates in the first image:

$$\begin{bmatrix} x_{\text{img1}} \\ y_{\text{img1}} \\ 1 \end{bmatrix} = H \begin{bmatrix} x_{\text{img2}} \\ y_{\text{img2}} \\ 1 \end{bmatrix}.$$

Because real data may contain outliers or mismatches, I used a robust estimation procedure based on *RANdom SAmple Consensus* (RANSAC). RANSAC iterates over subsets of matches, computes a candidate H, and then scores how many total matches agree with that candidate (within a predefined reprojection error). The best-scoring H is retained. This procedure greatly reduces the impact of incorrect correspondences.

2.4 Warping and Blending

Once H is established, I *warped* the second image (img2) into the coordinate frame of the first image (img1) by applying:

$$\tilde{I}_2(\mathbf{u}) = I_2(H^{-1}\mathbf{u}),$$

for every pixel coordinate $\mathbf{u}$ in the output canvas. In practice, I computed the bounding box of the warped image and create a larger canvas that can accommodate both img1 and the warped img2. Any negative coordinates are translated back into the canvas to ensure a seamless result.

I next *blended* the overlapping regions by a simple overwrite or mask-based approach, although more sophisticated strategies (e.g., multi-band blending) can be applied to achieve smoother transitions.

2.5 Implementation Details

All components are implemented in Python using the OpenCV library (version 4.x). The pipeline steps are coded as follows:

1. **Preprocessing:** I converted inputs to grayscale and optionally resize for faster processing.
2. **Feature Extraction:** I used `cv2.SIFT_create()` to get keypoints and descriptors.
3. **Matching and Filtering:** `BFMatcher().knnMatch()` is used for the KNN matches, followed by Lowe's ratio test.
4. **Homography Estimation:** `cv2.findHomography(..., RANSAC)`.
5. **Warping and Composition:** `cv2.warpPerspective(...)` for warping, then a masking approach for merging images.

Throughout our experiments, I used a ratio threshold $\alpha = 0.75$ and a RANSAC reprojection threshold of 5.0 pixels.

3. Experimental Results

3.1 Simple Translation on the X Axis

To establish a baseline, I first evaluated our pipeline on two images that differ by an approximate horizontal shift. Figure 1 shows an example pair (the *Taj Mahal* photos) with the detected SIFT keypoints overlaid on each image. Despite the uniform background and relatively repeated structure, SIFT was able to find several reliable features, especially around architectural details.

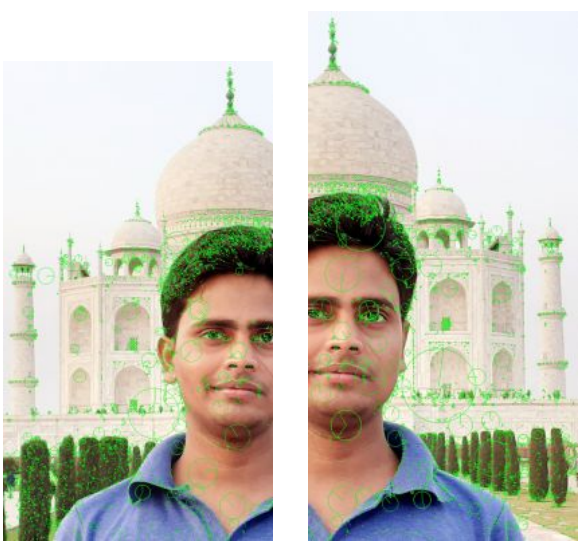

Figure 1: Detected SIFT keypoints for the two input images in the simple translation scenario.

Next, Figure 2 illustrates the good matches that survived Lowe's ratio test. I observed tight clustering of matches around the dome and minarets, which are distinctive features.

Figure 2: Good keypoint matches after ratio test (lines connect corresponding features).

Using these matches, a homography was computed to warp the second image into the first image's coordinate space. The final mosaic is shown in Figure 3, successfully aligning the main architectural elements. Note that a straight vertical seam is visible in the sky region, which could be improved with advanced blending methods.

Figure 3: Stitched panorama under pure translation. The images align well, with a minor seam visible in the sky.

3.2 Scaling and Translation Combined

Next, I tested scenarios where one image was also scaled relative to the other, in addition to being translated. I used a set of AI-generated cityscape images (Figure 4), where one image was manually rescaled by before stitching.

Figure 4 displays the detected SIFT keypoints for both images. Notice that although certain buildings appear more (or less) prominent in the scaled version, there remain enough common structures for robust matching.

Figure 4: Keypoints in the scaled cityscape images. Buildings and debris provide strong, repeatable features.

In Figure 5, I showed the final good matches. Many correspondences cluster around distinctive vertical edges, windows, and rubble in the scene. After applying RANSAC to remove outliers, the resulting homography accurately compensates for both translation and scale differences.

Figure 5: Good matches between scaled images in the cityscape scenario.

Figure 6 shows the stitched panorama, where the scaled image has been warped to match the reference image. The alignment is consistent across building edges, and the overall mosaic has few visible seams aside from brightness differences in certain regions.

Figure 6: Stitched panorama of the cityscape under combined scale + translation. Note the proper alignment of large building facades.

3.3 Black-and-White Film Photos (Real-World Test)

Finally, we experimented with a set of *black-and-white analog photos* (Figure 7). These images had noticeable scratches and variable contrast, providing a test for the robustness of our feature matching approach.

Figure 7: Black-and-white film images with SIFT keypoints. Imperfections like scratches pose additional challenges.

Despite the film artifacts, enough corner-like features were detected on building edges and the river bank to yield strong matches (Figure 8). Figure 9 showcases the final stitched panorama. The alignment is satisfactory, although minor perspective distortions arise because the buildings were not entirely planar and the shots were taken from slightly different angles.

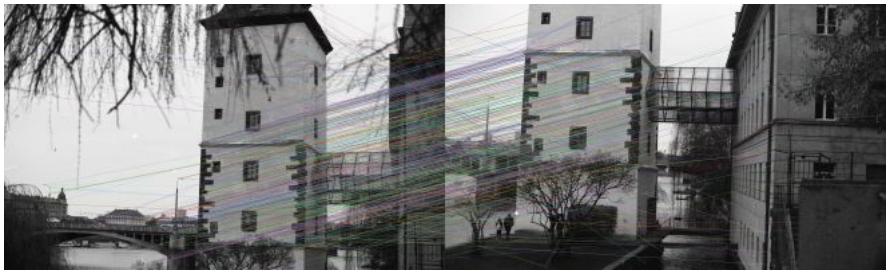

Figure 8: Match visualization for the black-and-white film images.

Figure 9: Stitched result for the black-and-white film photos. Alignment is broadly accurate, confirming the pipeline's robustness to scratches and contrast variations.

4. Conclusions and Future Work

The approach used has proven effective in dealing with translations, scaling, and scene noise in both synthetic and real-world images, including damaged black-and-white film. RANSAC plays a critical role in ensuring stability by discarding outlier matches, which helps maintain alignment among repetitive structures and minor viewpoint changes.

However, the method has limitations, particularly in scenes with significant parallax, where it can introduce bias due to its reliance on a single homography. Also, the basic pixel-overwriting blending strategy can create visible seams or brightness inconsistencies. In addition, while SIFT is robust, it can be slow compared to alternatives such as ORB, posing a challenge for large-scale or real-time applications.

Future enhancements may include the use of piecewise homographies or optical flow to better handle strong parallax, advanced blending techniques to smooth seams, and faster keypoint detectors to improve performance. Extending the system to support multi-image stitching could enable the creation of comprehensive 360-degree panoramas. These enhancements would increase the system's flexibility and ability to handle more complex scenarios while maintaining its robust core processes.